

Supplementary Material: Same Product, Different Visibility

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A Additional Experimental Detail

A.1 Representation audit

Each primary representation preserves the title and all raw attribute key-value atoms. C0 uses source order, C1 reverses attributes, and C2s1–C2s5 use independent deterministic per-product permutations. The audit checks atom multiplicity, character and token multisets, numeric substrings, and source JSON values. These tests isolate ordering within the supported equivalence definition; they do not assert that all strings are equally fluent.

A.2 Additional mitigation variants

Using the same development protocol, late mean, late maximum, and late top-3 aggregation all satisfy the numerical invariance tolerance. On held-out WANDS their cNDCG@10 values are 0.8058, 0.8093, and 0.8106; each paired confidence interval establishes a loss relative to 0.8247 Original. On ESCI they achieve 0.7026, 0.7041, and 0.7026, with all intervals including zero. These variants are omitted from the main-paper frontier plot for legibility; their paired intervals appear in the main-paper effectiveness-interval figure.

A.3 Official ESCI pools

Official judged pools contain a median of 16 products. MiniLM, BGE, and GTE obtain NDCG@10 of 0.6853, 0.7124, and 0.7168, and micro VI@10 of 7.60%, 6.65%, and 11.55%. At top 20, micro VI is 3.00%, 2.86%, and 5.89%. Query-macro VI@20 is 1.30%, 1.30%, and 2.88%.

A.4 Scope of the query taxonomy

Exploratory query classes were assigned by deterministic lexical rules. Because they lack blinded human annotation and agreement measurement, we do not use them to support the main causal claim. Query-level outputs retain the identifiers required for a later annotation study.

A.5 Result provenance

All values in the paper map to machine-readable source files in RESULT_PROVENANCE.md. The audit and output tests can be run with:

```
phase2/.venv/Scripts/python.exe -m pytest
phase3/tests -q
```

B Detailed Catalog-Wide Sensitivity

Table S1: Sensitivity among highest-relevance pairs. Micro and macro VI@20 are percentages; macro values include query-bootstrap 95% CIs. RR is median rank range.

Data	Retriever	RR	Micro	Macro VI@20
WANDS	MiniLM	207	13.26	19.17 [17.09,21.27]
	BGE-base	81	9.41	12.05 [10.58,13.67]
	GTE	107	11.33	14.43 [12.94,15.98]
	E-commerce	149	6.07	8.46 [6.99,10.08]
ESCI	MiniLM	2	4.67	4.04 [3.19,4.99]
	BGE-base	2	4.42	3.66 [2.84,4.57]
	GTE	3	8.25	6.91 [5.78,8.13]
	E-commerce	10	7.22	6.18 [5.17,7.30]

C Target-Only Primary Permutations

Each target is independently inserted against the same C0 competitors. This analysis was added after the confirmatory study and is not a new preregistered test. C0 scores and target ranks exactly reproduce the frozen outputs. Positive stored rank deltas denote a worse rank. All variant ranks and signed deltas, along with query-level VI, are released.

Table S2: Target-only VI@20 (percent), with query-bootstrap 95% CIs. All competitors remain in C0.

Data	Retriever	Micro	Macro (95% CI)
WANDS	MiniLM	12.72	18.96 [16.81,21.13]
WANDS	BGE-base	8.84	11.36 [9.88,12.96]
WANDS	GTE	11.39	14.60 [13.05,16.20]
ESCI	MiniLM	3.11	2.91 [2.14,3.78]
ESCI	BGE-base	3.16	2.42 [1.77,3.16]
ESCI	GTE	6.68	5.88 [4.83,7.03]